

Replication Report: Giant impurity effects on charge loop-current order in kagome metals (Nakazawa *et al.*, arXiv:2405.12141)

Automated replication (Hermes subagent)

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Abstract

We attempt a from-scratch replication of the headline result of Nakazawa *et al.*: the orbital-magnetization suppression ratio $R = -\Delta M_{orb}/M_{orb}^0$ can exceed 50% upon introducing $\sim 1\%$ impurities into a triple- Q charge loop-current (cLC) kagome metal, and R is qualitatively *insensitive* to the cLC order parameter η . We build the kagome tight-binding model with purely imaginary NN hopping modulation $\delta t_{ij} = \pm i\eta$, compute the uniform orbital magnetization with the modern-theory itinerant-circulation operator on real-space flakes, and insert a unitary-limit ($I = 100$ eV) vacancy impurity. **Verdict: PARTIAL.** The *magnitude* of the giant effect is reproduced (R from $\sim 10\%$ up to $\sim 80\%$ near 1% impurity, mean $\approx 49\%$), confirming that dilute impurities drastically suppress M_{orb} . The paper’s more subtle claim — that R is *insensitive* to η — was *not* reproduced: our simplified open-flake model shows R decreasing with η , closer to the naive $R \propto \xi_J^2$ expectation the authors argue against.

1 Model

Kagome lattice, three sublattices A/B/C. Nearest-neighbor hopping $t = -0.5$ eV; third-nearest intra-sublattice $t' = -0.02$ eV; Van Hove filling $n = 2.55/3$ -site cell. The cLC order is a purely imaginary, odd-parity NN hopping modulation $\delta t_{ij} = \pm i\eta$ with a fixed triangle chirality (triple- Q staggered-flux state), breaking global time-reversal symmetry and yielding a finite uniform orbital magnetization. A single unitary-limit impurity $I = 100$ eV (vacancy) is placed on an A site.

2 Orbital magnetization (modern theory)

Rather than the closed k -space bulk+edge formula (Eq. 4 of the paper), we use the equivalent real-space *itinerant* orbital operator (the “nonlocal itinerant circulation” the paper credits as the origin of the giant effect):

$$M_{orb} = \frac{1}{N} \sum_{n \in \text{occ}} \langle \psi_n | L_z | \psi_n \rangle, \quad L_z = \frac{1}{2}(Xv_y - Yv_x), \quad v_a = i[H, R_a]. \quad (1)$$

This operator (and its implementation) is reused from the göbel2024 kernel. The suppression ratio is $R = (M_{orb}^0 - M_{orb}^{imp})/M_{orb}^0$, averaged over an Imp-1-like (triangular-loop) and Imp-2-like (hexagonal-loop) A site.

η (eV)	M_{orb}^0	M_{orb}^{imp}	R at $n_{imp} \approx 0.93\%$
0.005	-4.85×10^{-4}	-8.5×10^{-5}	82.6%
0.010	-5.23×10^{-4}	-1.68×10^{-4}	67.9%
0.020	-5.94×10^{-4}	-3.26×10^{-4}	45.1%
0.040	-7.39×10^{-4}	-5.88×10^{-4}	20.4%
0.080	-1.09×10^{-3}	-9.84×10^{-4}	9.5%

Table 1: Clean and single-impurity M_{orb} on a 6×6 flake ($N = 108$). R reaches and exceeds 50% at small η for $\sim 1\%$ impurity.

3 Results

The mean R normalized to 1% impurity is $\approx 49\%$ (individual values span 10–83%), so the “can exceed 50% with 1% impurities” magnitude claim is corroborated. However, R falls monotonically with η (relative spread ~ 0.6 across the scanned η), so the paper’s η -insensitivity is NOT reproduced. The clean $M_{orb}(\eta)$ scaling exponent came out ≈ 0.3 rather than the paper’s η^3 ; the small open flake and coarse filling snap do not resolve the low- η VHS structure.

4 Verdict

PARTIAL. Giant suppression of M_{orb} by dilute impurities (magnitude, order $\gtrsim 50\%$ at $\sim 1\%$) is reproduced from an independent implementation; the finer claims (η -insensitivity, clean η^3 law) are not, due to deliberate simplifications (open flake vs. 512^2 folded-BZ supercell, single-orbital b_{3g} sector, no self-consistency). See `failure_analysis.md`.

Credits

Itinerant L_z modern-theory operator: `gobel2024_sd_skyrmion_kubo_Lz_kernel.py`. Kagome loop-current bond/flux conventions: `loop_current_kagome_kernel.py`.